KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

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#1.4

ODE Closed form solutions

- Separable Equations
- Integrating factor
- Linear Homogeneous Equations with Constant Coefficients
- Linear Nonhomogeneous Equations with Constant Coefficients
- Power Series Method
- Laplace Transform
- Bernoulli Equation
- First-Order Constant Coefficient Systems
- Matrix exponential
- Q & A

Closed-Form Solutions for ODEs

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- **Limitation:** Not all ODEs have closed-form solutions. Many require numerical methods.

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  - Example: $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$, solution: $x^2y + xy^2 = C$

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  - Example: $y'' - 3y' + 2y = e^x$, solution: $y = C_1 e^x + C_2 e^{2x} - xe^x$

Method of Undetermined Coefficients

- **Purpose:** Find a particular solution y_p for nonhomogeneous linear ODEs with constant coefficients.
- **Form:** $ay'' + by' + cy = f(x)$, where $f(x)$ is a polynomial, exponential, sine, cosine, or a combination.
- **Method:**
  1. Determine the form of y_p based on $f(x)$.
  2. Substitute y_p into the ODE and solve for the unknown coefficients.
  3. If y_p overlaps with the homogeneous solution y_h, multiply y_p by x (or x^2, etc.) until there is no overlap.
- **Example:** $y'' - 3y' + 2y = 3e^{2x}$
  - $y_h = C_1 e^x + C_2 e^{2x}$
  - Assume $y_p = Axe^{2x}$ (since e^{2x} is in y_h)
  - $y_p' = Ae^{2x} + 2Axe^{2x}$, $y_p'' = 4Ae^{2x} + 4Axe^{2x}$
  - Substitute into the ODE: $(4Ae^{2x} + 4Axe^{2x}) - 3(Ae^{2x} + 2Axe^{2x}) + 2Axe^{2x} = 3e^{2x}$
  - Simplify: $Ae^{2x} = 3e^{2x}$, so $A = 3$
  - $y_p = 3xe^{2x}$
  - General solution: $y = C_1 e^x + C_2 e^{2x} + 3xe^{2x}$

Variation of Parameters

- **Purpose:** Find a particular solution y_p for nonhomogeneous linear ODEs with constant coefficients or variable coefficients.
- **Form:** $y'' + p(x)y' + q(x)y = f(x)$
- **Method:**
  1. Find the homogeneous solution $y_h = C_1 y_1(x) + C_2 y_2(x)$.
  2. Compute the Wronskian: $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$.
  3. Find $u_1(x) = -\int \frac{y_2(x) f(x)}{W(y_1, y_2)} dx$ and $u_2(x) = \int \frac{y_1(x) f(x)}{W(y_1, y_2)} dx$.
  4. $y_p = u_1(x) y_1(x) + u_2(x) y_2(x)$.
- **Example:** $y'' + y = \sec(x)$
  - $y_h = C_1 \cos(x) + C_2 \sin(x)$
  - $W(\cos(x), \sin(x)) = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix} = \cos^2(x) + \sin^2(x) = 1$
  - $u_1(x) = -\int \frac{\sin(x)\sec(x)}{1} dx = -\int \tan(x) dx = \ln|\cos(x)|$
  - $u_2(x) = \int \frac{\cos(x)\sec(x)}{1} dx = \int 1 dx = x$
  - $y_p = \cos(x) \ln|\cos(x)| + x \sin(x)$
  - General solution:
    $y = C_1 \cos(x) + C_2 \sin(x) + \cos(x) \ln|\cos(x)| + x \sin(x)$

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- **Disadvantages:** The resulting power series may have a limited radius of convergence.

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  3. Substitute into the ODE: $\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$.

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  8. $y(x) = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$.

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  9. $y(x) = a_0 \left(1 + \frac{x^3}{6} + \frac{x^6}{180} + \cdots\right) + a_1 \left(x + \frac{x^4}{12} + \frac{x^7}{504} + \cdots\right)$.

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  - Translation in t: $\mathcal{L}\{u(t-a)f(t-a)\} = e^{-as}F(s)$ where $u(t)$ is the Heaviside step function.

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  - **Convolution Theorem:** If $F(s) = G(s)H(s)$, then $f(t) = \mathcal{L}^{-1}\{F(s)\} = (g * h)(t) = \int_0^t g(\tau)h(t-\tau)d\tau$.

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  - **Complex Inversion Formula (Bromwich Integral):**

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{st} F(s)\, ds$$

where c is a real number greater than the real part of all singularities of $F(s)$. (Used in advanced cases.)

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  - Partial fraction decomposition: $\frac{3s+1}{(s+3)(s-2)}=\frac{A}{s+3}+\frac{B}{s-2}$.
  - Solve for A and B: $3s+1=A(s-2)+B(s+3)$.
  - Setting $s=2$, we get $7=5B$, so $B=\frac{7}{5}$.
  - Setting $s=-3$, we get $-8=-5A$, so $A=\frac{8}{5}$.
  - $\mathcal{L}^{-1}\left\{\frac{3s+1}{s^2+s-6}\right\}=\mathcal{L}^{-1}\left\{\frac{8/5}{s+3}+\frac{7/5}{s-2}\right\}=\frac{8}{5}e^{-3t}+\frac{7}{5}e^{2t}$.

Solving ODEs using Laplace Transform

- **Procedure:**
  1. Take the Laplace transform of the entire ODE.

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  1. Take the Laplace transform of the entire ODE.
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- **Advantages:**
  - Handles initial conditions naturally.
  - Simplifies solving nonhomogeneous equations with piecewise continuous forcing functions.

Laplace Transform: Example

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  6. Take the inverse Laplace transform:
     $y(t) = \cos(2t) + \frac{3}{10}\sin(2t) - \frac{1}{5}\sin(3t)$.

Bernoulli Equation: Introduction

- **Definition:** A Bernoulli equation is a first-order nonlinear ordinary differential equation of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

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  - For other values of n, the equation is nonlinear.
- **Goal:** Transform the Bernoulli equation into a linear equation, which can be solved using an integrating factor.

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First-Order Constant Coefficient Systems: Introduction

- **Form:** A system of first-order linear ODEs with constant coefficients can be written as:

$$\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$$

where $\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$, $\mathbf{x}'(t) = \begin{pmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{pmatrix}$, and $\mathbf{A}$ is an $n \times n$ constant matrix.

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$$\begin{cases} x_1'(t) = ax_1(t) + bx_2(t) \\ x_2'(t) = cx_1(t) + dx_2(t) \end{cases} \quad \Longrightarrow \quad \begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

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- **Goal:** Find the general solution $\mathbf{x}(t)$ that satisfies the system.

Solving Systems: Eigenvalues and Eigenvectors

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Case 1: Distinct Real Eigenvalues

- **If A** has n distinct real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ with corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$,

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- **Solution:** $\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$.

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- **Generalized Eigenvectors:** Solve $(\mathbf{A} - \lambda \mathbf{I})\mathbf{w} = \mathbf{v}$, where $\mathbf{v}$ is an eigenvector.
- **Solutions:** $\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$, $\mathbf{x}_2(t) = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$, etc.

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- Distinct real eigenvalues lead to simple exponential solutions.
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- The general solution is a linear combination of the fundamental solutions.

Matrix Exponential: Definition and Solution

- **Matrix Exponential:** For an $n \times n$ matrix $\mathbf{A}$, the matrix exponential $e^{\mathbf{A}t}$ is defined as:

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!}$$

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- **Advantages:**
  - Provides a compact and elegant representation of the solution.
  - Useful for theoretical analysis and understanding the behavior of the system.
  - Can be computed using various methods, including eigenvalue decomposition.

Matrix Exponential: Computation and Example

- **Computation using Eigenvalues and Eigenvectors:**
  - If $\mathbf{A}$ is diagonalizable, i.e., $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, where $\mathbf{D}$ is a diagonal matrix of eigenvalues and $\mathbf{P}$ is a matrix of eigenvectors, then:

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  - $e^{\mathbf{A}t} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} 1/2 & 1/4 \\ 1/2 & -1/4 \end{pmatrix} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t} - e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}$.

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  - $e^{\mathbf{A}t} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} 1/2 & 1/4 \\ 1/2 & -1/4 \end{pmatrix} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t} - e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}$.
  - $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} \\ e^{3t} - e^{-t} \end{pmatrix}$.

Nonhomogeneous Systems: Introduction

- **Form:** A nonhomogeneous system of first-order linear ODEs with constant coefficients is given by:

$$\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{f}(t)$$

where $\mathbf{x}(t)$ is the vector of unknown functions, $\mathbf{A}$ is a constant matrix, and $\mathbf{f}(t)$ is a vector of nonhomogeneous terms.

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- **Goal:** Find the general solution $\mathbf{x}(t)$ that satisfies the nonhomogeneous system.
- **General Solution:** The general solution is the sum of the homogeneous solution $\mathbf{x}_h(t)$ and a particular solution $\mathbf{x}_p(t)$:

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  4. If any term in the guess for $\mathbf{x}_p(t)$ is also a solution to the homogeneous equation, multiply that term by t (or t^2, etc.) until there is no overlap.
- **Example:** If $\mathbf{f}(t) = \begin{pmatrix} 2e^t \\ t \end{pmatrix}$, guess $\mathbf{x}_p(t) = \begin{pmatrix} a_1 e^t + b_1 t + c_1 \\ a_2 e^t + b_2 t + c_2 \end{pmatrix}$.

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- **Fundamental Matrix:** (t) is a matrix whose columns are linearly independent solutions of the homogeneous system.
- **Example:** For a 2×2 system, if $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are linearly independent solutions, then $(t) = \begin{pmatrix} \mathbf{x}_1(t) & \mathbf{x}_2(t) \end{pmatrix}$.

Nonhomogeneous Systems: Example 1 (Undetermined Coefficients)

- **Problem:** Solve $\mathbf{x}'(t) = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \mathbf{x}(t) + \begin{pmatrix} 4e^{2t} \\ -2e^{2t} \end{pmatrix}$.

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Nonhomogeneous Systems: Example 2 (Variation of Parameters)

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Nonhomogeneous Systems: Matrix Exponential Solution

- **Form:** Consider the nonhomogeneous linear system:

$$\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{f}(t)$$

where $\mathbf{A}$ is a constant $n \times n$ matrix, $\mathbf{x}(t)$ is the vector of unknown functions, and $\mathbf{f}(t)$ is a vector of nonhomogeneous terms.

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  - $\int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{f}(\tau)\, d\tau$ represents the particular solution.
- **Advantage:** This method provides a compact and general formula for the solution of nonhomogeneous systems.

Matrix Exponential: Example and Computation

- **Example:** Solve the system:

$$\mathbf{x}'(t) = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}(t) + \begin{pmatrix} e^t \\ 0 \end{pmatrix}$$

with initial condition $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

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$$e^{\mathbf{A}t} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t} - e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}$$

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- **Simplify and Integrate:** After performing the matrix multiplication and integration, substitute the result into the general solution formula:

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- **Result:** Obtain the final solution $\mathbf{x}(t)$.

Homogeneous Systems with Variable Coefficients: Matrix Exponential (Formal)

- **Form:** Consider the homogeneous system:

$$\mathbf{x}'(t) = \mathbf{P}(t)\mathbf{x}(t)$$

where $\mathbf{P}(t)$ is a matrix with variable coefficients.

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- **Fundamental Matrix and Matrix Exponential Connection:** If $\mathbf{P}(t)$ commutes with its integral, i.e.,

$$\mathbf{P}(t)\int_{t_0}^{t}\mathbf{P}(\tau)d\tau = \left(\int_{t_0}^{t}\mathbf{P}(\tau)d\tau\right)\mathbf{P}(t)$$

then a fundamental matrix can be expressed as:

$$(t) = \exp\left(\int_{t_0}^{t}\mathbf{P}(\tau)d\tau\right)$$

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- **General Solution:** The general solution is then:

$$\mathbf{x}(t) = \exp\left(\int_{t_0}^{t}\mathbf{P}(\tau)d\tau\right)\mathbf{c}$$

where $\mathbf{c}$ is a constant vector determined by initial conditions.

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- **Important Note:** The commutativity condition is crucial. Without it, this form is generally incorrect.

Matrix Exponential: Limitation and Example

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$$\int_0^t \mathbf{P}(\tau)d\tau = \int_0^t \begin{pmatrix} \tau & 0 \\ 0 & 2\tau \end{pmatrix} d\tau = \begin{pmatrix} t^2/2 & 0 \\ 0 & t^2 \end{pmatrix}$$

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- **Check Commutativity:**

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Q & A